

Experts opinion on Budget 2020 for EVs

"From a customer's perspective, right now, cost of acquisition is very high. Penetration is only in public transport. That is also a subsidy-based transportation promoted by the government. There are certain schemes like FAME that have been provided, hence penetration level is high. But this particular thing will take time. As of now, government initiatives and expectations are more focused on larger buses. We are not satisfied with the budget apart from the increased customs and import duties. We think certain incentives need to be provided in this sector, which would boost primarily EVs, and entire auto industry."

—**Chandrakant Thakur, deputy general manager - CVD sales, Force Motors**

"Union Budget 2020 covered most expectations from different sectors; however, EVs did not get much mention in the speech. It would have been a major step if FIs and banks could get a little push to providing loans for EVs on lower interest rates. This could have motivated buyers to opt for EVs. Though the automobile industry did go through a little slow down in 2019-2020, the EV industry got a comparatively better response compared to 2018-2019. Electric two-wheelers witnessed more acceptance over electric four-wheelers. As infrastructure strengthens and awareness spreads, demand will only grow."

"Now, if we carefully look at the two-wheeler segment, the market leader is Honda followed by TVS, Bajaj, Yamaha and more. And for three-wheelers, Bajaj Auto is the market leader, then Piaggio, M&M, Atul Auto and TVS. For four-wheelers, Hyundai, M&M, Tata Motors and Honda Cars are the leaders. In terms of percentage, there are 81 per cent two-wheelers, four per cent three-wheelers, three per cent commercial vehicles and twelve per cent passenger vehicles."

—**Jeetender Sharma, founder and managing director, Okinawa**

"Concerning government policies, GST has been reduced on EVs to five per cent, which is an encouraging sign. Though the government has introduced FAME-II, we see very few companies qualified for this. The guiding principle behind FAME-II is to give a push to EVs in public transport and to encourage adoption of EVs by way of market creation and demand aggregation. In terms of budget, we want reduction of GST on lithium batteries to twelve per cent or five per cent as against eighteen per cent. This will help EV manufacturers to save on input costs."

—**Nishchal Chaudhary, founder and chief executive officer, BattRE**

"We expect from the government to cut down GST on components like batteries, chargers, etc, in addition to special incentive schemes for component suppliers for smooth supply chain management in India, financial support to R&D, minimum homologation and testing cost, free parking lots for EV users, special support to dealers of EV, and easy documentation system for EV registration. These can help boost the market share further. Currently, the market share of electric two-wheelers is seventeen per cent, whereas market share of electric three-wheelers (including e-rickshaws) is 82 per cent and of electric four-wheelers is one per cent."

—**Pankaj Tiwari, business development head, Nexzu Mobility**

"Retail and finance are some of the major challenges. Finance, which a consumer takes in to buy a vehicle, is not available easily, particularly for lower speed ranges that are 250W motor or less. The entire automotive industry is not happy with the budget. From the government's end, we would love to see more sophisticated infrastructure for making people aware of how electric products can be used and support for the existing companies to establish more infrastructure, in India particularly."

—**Sarthak Das, marketing manager, OMJAY EV Ltd**

in the world. Though there will be lesser greenhouse gas emissions, these will still protect the ozone layer and lessen carbon footprint.

Electric vehicles (EVs) are said to offer several benefits such as low emissions, high energy efficiency and low maintenance. These vehicles are also high in performance as their motors are smooth and quiet, and require less maintenance than those with internal combustion engines (ICEs).

However, some experts challenge this statement because to them EVs are not high-performance vehicles, as these cannot move smoothly on highways, mountains and terrains, and also require high cost of maintenance because these can be repaired only in authorised service centres, and garage car mechanics are not familiar with their motors and parts. Hence, unless these vehicles become more popular and almost completely take over conventional vehicles, it will be difficult to judge their full potential.

Another uniqueness of EVs is that these are overall digitally connected with all nearby charging stations, which permits finding a nearby charging station with the help of an app.

EVs also cost less to run because there is no gas to buy, no oil changes, no smog tests and fewer moving parts to break or wear out. Most importantly, if electric power is sourced from renewable resources, volume of greenhouse emissions will reduce further.

Challenges for EV market in India

Even though the EV sector promises lots of advantages, it is facing bundles of impediments in its way. When it comes to technology challenges, EVs have shorter range than gas-powered and conventional vehicles. EVs at this moment might give pain on long trips. A full recharge of the battery

pack might take eight hours, while fast-charging stations can also take up to thirty minutes to charge eight per cent (approximately). EV drivers have to plan carefully, because if their car battery runs out of power, there cannot be a quick solution.

To make EVs a success on Indian roads, several obstacles like inadequate charging infrastructure, reliance on battery imports, high price of EVs at present, inadequate electricity supply in parts of India, lack of quality maintenance and repair options have to be taken care of. Nonetheless, if technologies supporting the growth of EVs and batteries would pick up more, major obstacles such as high cost, limited

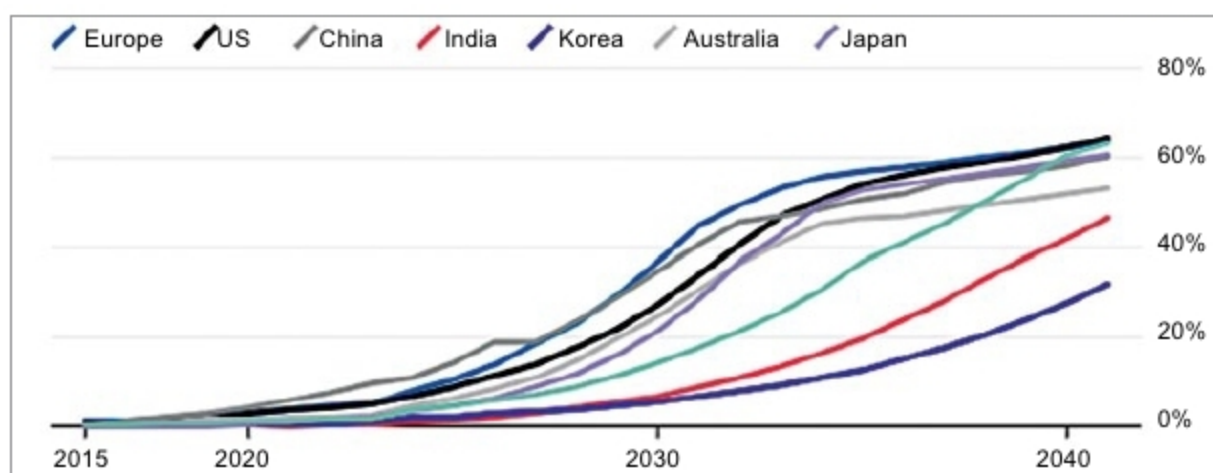


Fig. 1: Electric vehicles' sales penetration is expected to rise sharply after 2030 (Credit: Bloomberg New Energy Finance)